New Medical Technology and Clinical Practice: Evidence from AI Stroke Triage

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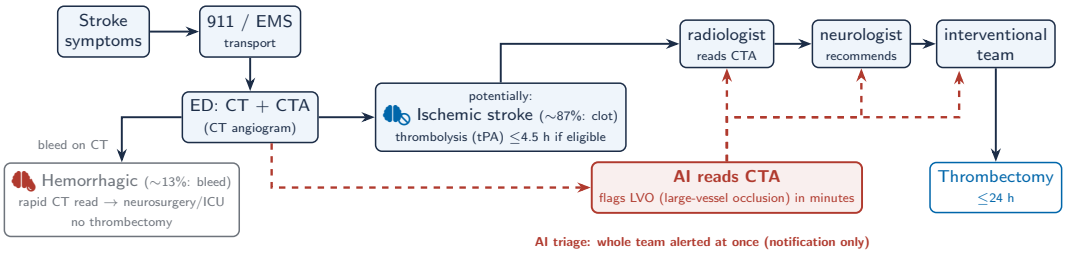
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Washington Health Economics Symposium, U.S. Capitol, 2026.10.5

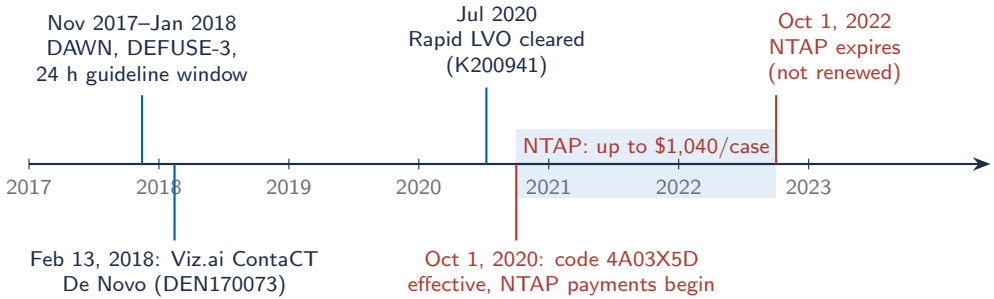
The problem: the best stroke treatment may not reach patients

- **Stroke**: leading cause of long-term disability; 795,000+ US strokes, $56B/yr (CDC)
- Mechanical **thrombectomy** is one of the most effective procedures for ischemic stroke: treat <3 patients, prevent one lifetime of disability (HERMES trials; *Lancet* 2016)
- Yet most eligible stroke patients never receive it; bottleneck: recognition speed



A temporary add-on payment made AI adoption visible

- Medicare: dedicated billing code + temporary **New Technology Add-on Payment (NTAP)**; the code gives a first national picture of clinical AI adoption
  - temporary top-up to the MS-DRG bundle for new inpatient technologies



▸ NTAP mechanics

This paper: does hospital AI stroke triage adoption change treatment?

Staggered DiD: within-facility changes in thrombectomy in early vs. late AI adopters

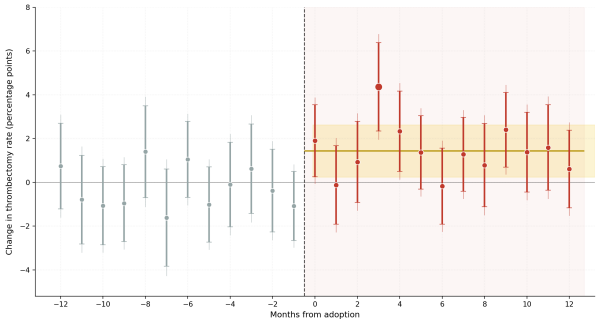
- National multi-payer claims: 420,294 ischemic stroke admissions, 2,828 hospitals, 2017–2023; 643 adopters (commercial, Medicare Adv., managed Medicaid; no FFS Medicare)
- AI adoption = first admission billing the AI code (ICD-10-PCS 4A03X5D)

  code descriptor: "Measurement of Arterial Flow, Intracranial, External Approach"
- Main restriction: facilities regularly treating ischemic stroke ($\geq$1 admission in 12+ of 84 study months); sporadic AI billers ($<$5 coded admissions) excluded; robust to alternatives

▸ sample flow ▸ billing-code validity

Finding 1: treatment rises at adoption, and stays up

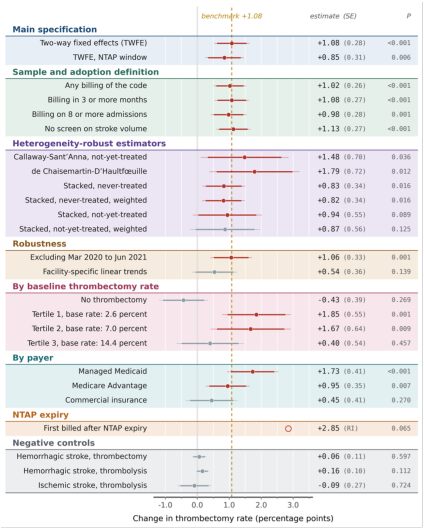
- **+1.08 pp** (95% CI 0.54, 1.62) on a 4.73% base ≈ **+23%**
- Flat before adoption; jump at adoption month
- No evidence of decay through two years



Event study around adoption (paper Fig. 3)

▸ two years

The estimate is robust to many alternative specifications/samples



- Alternative adoption defn: +0.98 to +1.13
- Robust estimators bracket the conservative TWFE benchmark: CSDID +1.48, dCDH +1.79, stacked +0.82–0.94
- Negative control: hemorrhagic stroke; thrombolysis null

▸ estimator overlay

Finding 2: gains where capability exists but is not yet saturated

Effect on thrombectomy use by baseline capability (2017–2019 thrombectomy rate):

Baseline group	Change (pp)	95% CI	Base rate
No thrombectomy	−0.43	(−1.20, 0.33)	0.0
Tertile 1	**+1.85**	(0.76, 2.93)	2.6
Tertile 2	**+1.67**	(0.42, 2.92)	7.0
Tertile 3	+0.40	(−0.66, 1.47)	14.4

1,179 facilities with $\geq$ 30 ischemic stroke admissions in 2017-2019. Of these, 645 performed no thrombectomy; the remaining 534 are split into equal tertiles by baseline rate. Difference across groups $P < 0.001$; SEs clustered by facility.

- Gains concentrate where a program exists but is **not yet saturated**; flat with no program or near saturation
- Consistent w/ **faster recognition** inside existing programs, not expansion

Finding 3: largest gains for publicly insured patients

Effect on thrombectomy use by payer (single interacted model):

Payer	Change (pp)	95% CI	Baseline %
Managed Medicaid	**+1.73**	(0.93, 2.52)	3.95
Medicare Advantage	+0.95	(0.25, 1.64)	4.42
Commercial	+0.45	(−0.35, 1.25)	5.53

Joint difference across payers $P = 0.048$. Admission-weighted; SEs clustered by facility.

- Largest gain on **lowest baseline**: AI narrowed a treatment gap within hospitals
- None of these payers received the add-on (FFS inpatient only), yet effects appear without it

Robustness & mechanism: what drives the treatment rise?

- **Not more scanning**: imaging utilization flat at adoption; volume growth predated adoption and shows up equally for imaging the software cannot read
- **Not substitution**: thrombolysis unchanged (-0.09; CI -0.62, 0.43)
- **Not due to transfers**: transfers to other acute-care hospitals unchanged
- **Not the subsidy mechanically**: no payer in our data received the add-on; 42 post-expiry adopters show a point estimate at least as large ($+2.85$, RI $P = 0.065$)

$\Rightarrow$ **faster in-window recognition of already-eligible patients**

▸ secondary outcomes

Transition: payment-linked AI billing code, fading w payment sunset

- The add-on expired Sep 2022 with no successor; the billing code (4A03X5D) remains on the books, yet billing dropped sharply once the payment stopped
- **No evidence of decay**: second-year average $+1.39$ pp, matching the first year; the 42 facilities adopting *after* expiry show a point estimate at least as large ($+2.85$; RI $P = 0.065$)
- Payment codes are also **measurement infrastructure**: when the payment lapsed, claims-based visibility of clinical AI use faded with it
- **Payment pathways** remain unsettled: a dedicated outpatient reimbursement mechanism for AI devices was proposed only in the CY2027 OPPS rule (SaMS; 91 FR 41734, Jul 2026), w/ ongoing legislative debate (e.g., Health Tech Investment Act)

Not established here: payment's causal effect on adoption, cost-effectiveness, FFS Medicare outcomes, or that permanent payment would do more.

▸ payment pathways ▸ billing-code validity

Takeaways

1. **What**: AI triage adoption raised stroke treatment ∼**23%**; persistent, robust, true
2. **So what**: within adopting hospitals, gains concentrate where baseline access was worst (managed Medicaid, unsaturated facilities): the technology narrowed gaps rather than widening them
3. **How**: a temporary payment created the adoption, the evidence, and the visibility; all three lapsed together (cf. Germany's DiGA: payment + evidence by design).



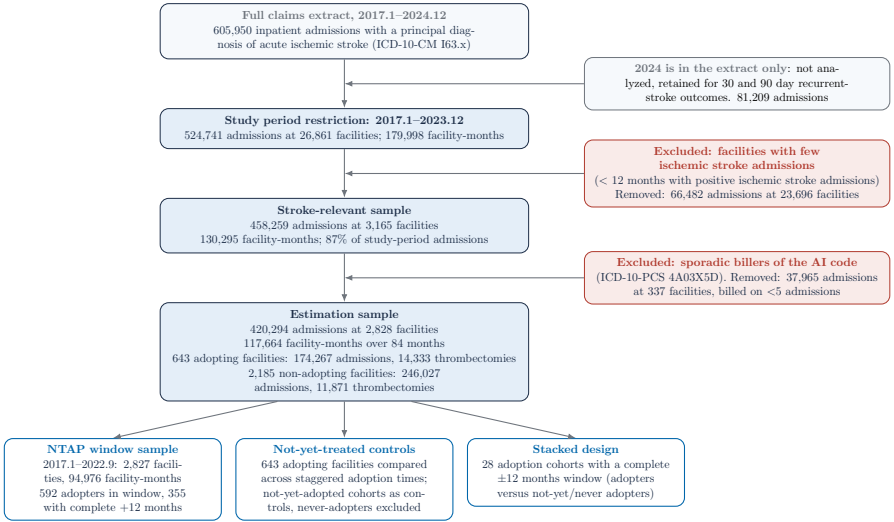
2026.10.29
World Stroke Day

B	E	F	A	S	T
Balance	Eyes	Face	Arm	Speech	Time: call 911

Mirror case, our 3D-printing paper: CPT codes but no payment attached — billing rare, ∼25–30% denied.

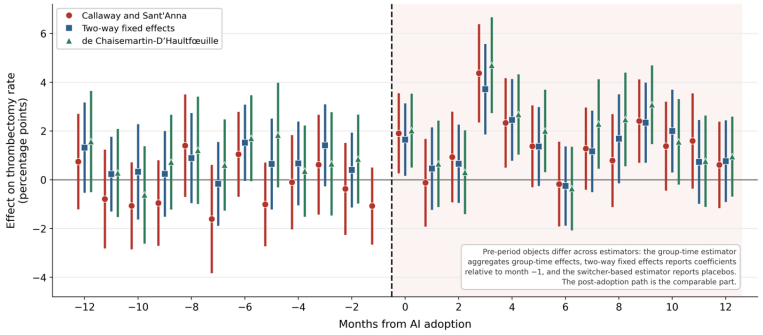
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Appendix: sample flow (paper Fig. 1)



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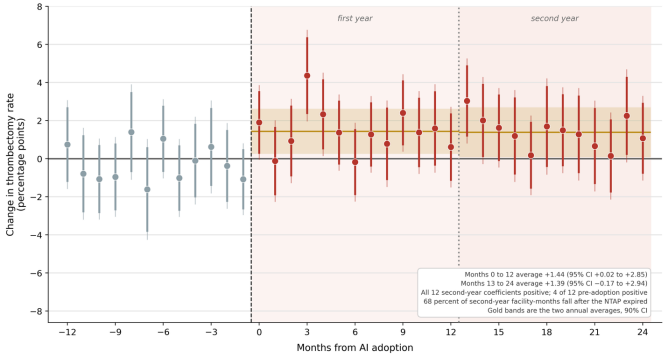
Appendix: estimator overlay (paper Fig. S4)



dCDH +1.79 / CSDID +1.48 / TWFE +1.08 / stacked +0.82–0.94: robust estimators bracket the conservative benchmark; post-adoption paths trace closely

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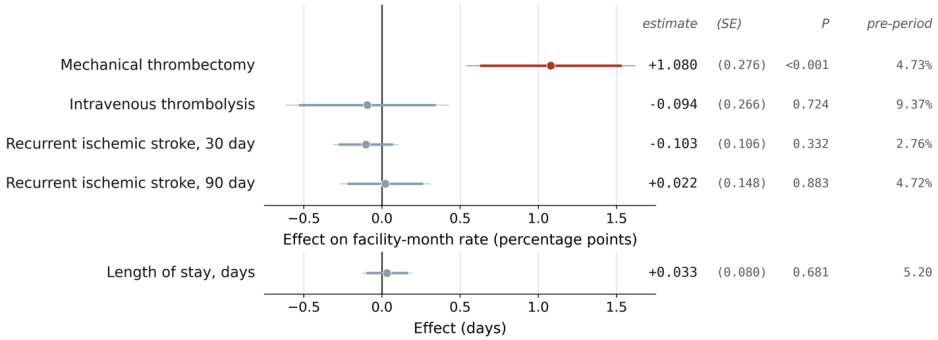
Appendix: two-year persistence (paper Fig. S2)



Months 0–12: +1.44 (CI 0.02, 2.85); months 13–24: +1.39 (CI −0.17, 2.94). No evidence of decay; not a claim of precise permanence

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Appendix: secondary outcomes and negative controls (paper Fig. S3)



Screened sample: 2,828 facilities, 117,664 facility-months. Thick bars 90% CI, thin lines 95% CI.

Thrombolysis −0.09: rules out >half substitution; hemorrhagic control +0.06 (null); recurrence/LOS: diluted nulls; mortality not observable.

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Appendix: magnitude arithmetic

- Adopters' 2023 admissions: 35,543 $\times$ +1.08 pp $\Rightarrow$ 384 added thrombectomies/yr (192–576)
- Trials' functional-independence margin (HERMES: 46% vs. 27%) $\Rightarrow$ illustratively $\approx$ 75 additional independent patients/yr (37–112)
- Computed on our adopters only; excludes traditional Medicare FFS. National figure larger under any reasonable extrapolation (not estimated)

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Appendix: NTAP mechanics and the billing code

- **NTAP**: temporary Medicare add-on, on top of the MS-DRG payment, for qualifying new technologies; inpatient FFS claims only; expires after ∼2–3 fiscal years by design
- **History**: created by Congress in 2000 (BIPA), payments since FY2002; early recipients incl. drotrecogin alfa (sepsis biologic, FY2003), later e.g. CAR-T cell therapies (FY2019)
- This AI triage software was the first **AI software** granted an NTAP (FY2021 IPPS final rule, Sep 2020)
- **The billing code** (ICD-10-PCS `4A03X5D`) is how hospitals claimed the add-on, and how we (and CMS) observe adoption in claims
- Design implication: the add-on paid only traditional Medicare FFS, so *none* of the payers in our data were subsidized; effects appear anyway

Appendix: billing-code validity, a conservative measure

- Billing is payment-sensitive: it declined after the add-on expired while thrombectomy stayed up; the code tracks *payment incentives*, not abandonment
- Unbilled use and left-censoring misclassify adopters as non-adopters $\Rightarrow$ bias **toward zero**; the main estimate is a **lower bound**
- Payment-window co-primary replicates: $+0.85$ (CI 0.25, 1.45)
- Four alternative sample/adoption definitions: 0.98–1.13

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Appendix: the payment-pathway landscape

- **Status quo**: NTAP is time-limited by design; after expiry there is **no systematic Medicare payment pathway** for clinical-AI services of this kind
- **Legislative debate**: proposals to create Medicare payment for algorithm-based health services (e.g., the Health Tech Investment Act discussion); a dedicated outpatient reimbursement mechanism proposed in the CY2027 OPPS rule
- **International contrast**: Germany's DiGA fast-track pairs provisional reimbursement w/ mandatory evidence generation; payment and evaluation designed together
- **Our evidence adds**: a time-limited add-on generated adoption, measurable practice change, *and* the data to evaluate it; the sunset-plus-evidence design worked once

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